The Market tab

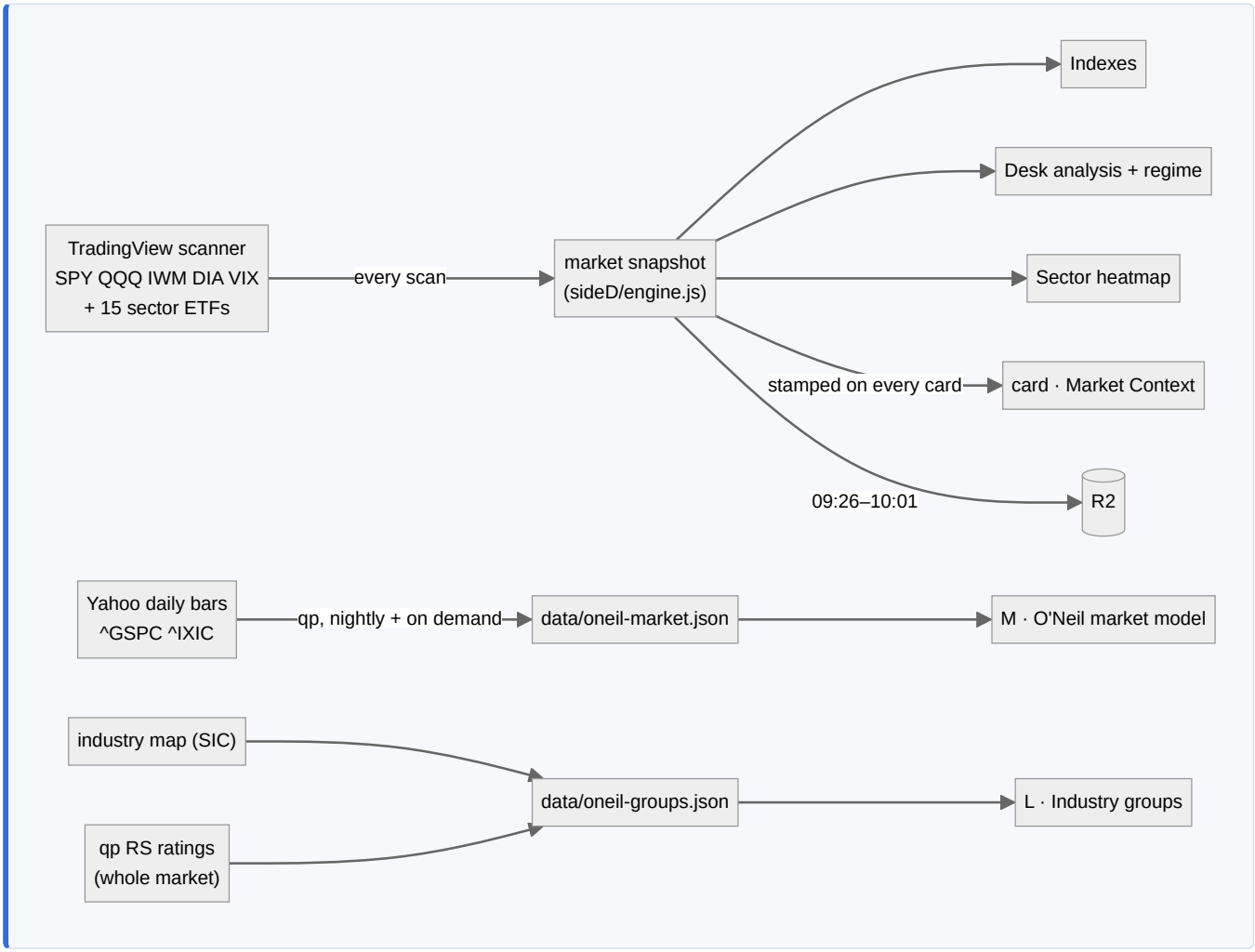
Every block on a tool's **Market** tab: where its numbers come from, how each verdict is computed, every threshold, and where the same numbers also land (each card's Market Context, and register R2). Written from `src/sideD/*.js`, `src/routes/market.js`, `quant-platform/chart/oneil.py`, `quant-platform/chart/groups.py` and the Market tab code in `public/index.html`.

Two systems sit on this tab, kept apart on purpose:

System	Built by	Answers	Blocks
The desk's read	each tool, on every scan	what the tape is doing right now	Indexes, Desk analysis, Sector heatmap
O'Neil's method	qp, after each close	should you be buying at all, and where the leadership is	M · market model, L · industry groups

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Where the data comes from



- **The snapshot** is built at the start of the market-context stage of each discovery scan (`buildMarketSnapshot`). The tab reads the latest one through `GET /api/market/snapshot`; if none exists yet it builds one on the spot.
- **TradingView columns asked for:** `close`, `change`, `change|1W`, `SMA5`, `SMA20`, `SMA50`, `SMA200`, `BB.upper`, `BB.lower`, `ADX`, `VWAP`, `relative_volume_10d_calc`.
- **No regime rather than a wrong one.** If SPY and QQQ are both missing any of close/SMA5/20/50/200, the scan records no regime (D-S2) and the cards keep no context.
- **O'Neil files** are published by qp (nightly at 22:30 UTC, and on request when missing). `GET /api/market/oneil` and `/api/market/groups` ask qp to build them when the file is absent.

Folds: **Indexes** and **M · O'Neil market model** open by default; the rest closed. Which folds are open is remembered in the browser.

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1. Indexes · live

Five tiles in order **SPY, QQQ, DIA, IWM, VIX**: price, change today, change over the week. Tapping a symbol opens its chart.

2. M · O'Neil market model

qp's reconstruction of O'Neil's market direction from the **S&P 500 (^GSPC)** and the **Nasdaq Composite (^IXIC)**, daily bars from Yahoo.

The rules (`oneil.py` THRESHOLDS)

Rule	Value	Meaning
Distribution day	close down **≥ 0.2 %** on volume **higher than the prior session**	institutions selling
Stalling day	heavy volume (above the prior session or the 25-day average), close in the **lower half** of the range, gain **under 0.2 %**	selling absorbed: also counts as distribution
Live window	**25 sessions**	a distribution day expires after 25 sessions…
Recovery	index **trades 5 % above** that day's close (intraday high)	…or when the index has moved 5 % past it
Under pressure	**5** live distribution days	
Correction	**6** live distribution days, or the uptrend's low broken	
Rally attempt, day 1	the session after the **lowest close** of the correction	a lower close restarts the count
Follow-through day	day **4 or later**, gain **≥ 1.7 %**, volume above the prior session	confirms a new uptrend; "late" after day 7, "very late" after day 11 (still counted)

Volume above the 50-day average is **reported** beside a follow-through but does not decide it.

Status

Each index is walked forward through its own state machine; the **published status is the worse of the two indexes**:

Status	Shown as	What O'Neil does (the tab's text)
`confirmed_uptrend`	Confirmed uptrend	Buy breakouts. This is the state the method is designed for.
`uptrend_under_pressure`	Uptrend under pressure	Stop initiating new positions; tighten stops on open ones. A name holding up through distribution is a leader candidate for the next follow-through.
`market_in_correction`	Market in correction (including a rally attempt with no follow-through yet)	No new buying. Three out of four stocks follow the market, and this one is going down.

What the block shows

- the status, its date, and a **because** line (e.g. “5 distribution days live on the Nasdaq Composite (5 is the threshold)”);
- **Live distribution** per index;
- the **Follow-through**: date, day number, gain, timing;
- **Since the FTD**: the number of sessions, with a band;
- the **Rally attempt**: day N off a dated low;
- the exposure guidance above;
- a **distribution-day table**: date, index, kind (distribution or stalling), move %, volume ratio, sessions until it expires;
- the rules in use, a caveat (roughly one follow-through in four fails), and the model note (a reconstruction from O’Neil’s published descriptions).

Also on each card: CANSLIM **M** (status, live distribution days, and how this stock behaved on those days). In R2: `oneilStatus`, `oneilDistributionDays`, `oneilSessionsSinceFtd`, `oneilFtdDate`.

3. Desk analysis

The desk’s own read, in three cards plus a final regime. Computed from SPY (QQQ when SPY is incomplete) unless noted.

Short-term bias (`computeMarketBiasDetail`)

Signal	+1	−1
SPY day	> +0.3 %	< −0.3 %
QQQ day	> +0.3 %	< −0.3 %
IWM day	> +0.3 %	< −0.3 %
VIX day	< −2 % (+1)	> +1 % (−1), > +3 % (−2)
SPY week	> +1 %	< −1 %
QQQ week	> +1 %	< −1 %

Sum ≥ **+3** BULLISH, ≤ **−3** BEARISH, otherwise NEUTRAL. The card shows the score and each signal.

Mid-term stage (`computeMarketStage`)

Six checks, counted as bull signals:

1. close > SMA5

2. close > SMA20
3. SMA5 > SMA20
4. SMA20 > SMA50
5. “1H close > 20H MA”. The code uses the daily close > SMA20 as a proxy, which is check 2 again.
6. “1H 5MA > 20H MA”. The code uses SMA5 > SMA20, which is check 3 again.

Checks 5 and 6 repeat 2 and 3, so the count runs on four distinct facts, two of them counted twice. See *D-D5*.

Stage	Rule
UPTREND	≥ 5 bull, or 4 with close > SMA5
PULLBACK	3–4 bull, close < SMA5, SMA5 > SMA20
REBOUND	2–3 bull, close > SMA5, SMA5 < SMA20
SIDEWAYS	2–3 bull otherwise
DOWNTREND	fewer

Bollinger position (close between the bands): ≥ 75 % `UPPER`, ≤ 25 % `LOWER`, else `MID`.

Long-term bias (`computeLongTermBias`)

	SMA50 > SMA200 (golden cross)	SMA50 < SMA200
close > SMA200	BULLISH	RECOVERING
close < SMA200	WEAKENING	BEARISH

Also shown: the distance from the 200-day in %.

Final regime (`computeRegime`)

Long-term × mid-term, then the Bollinger overlay:

Long \ Mid	UPTREND	PULLBACK	REBOUND	SIDEWAYS	DOWNTREND
BULLISH	STRONG_UP	PULLBACK_BULL	UP	CHOP_BULL	CORRECTION
RECOVERING	WEAK_UP	RECOVERY	RECOVERY	BASING	DOWN
WEAKENING	RECOVERY	TOPPING	BEAR_RALLY	CHOP_BEAR	DOWN
BEARISH	BEAR_RALLY	DOWN	BEAR_RALLY	BASING	STRONG_DOWN

STRONG_UP at the upper band becomes **EXTENDED_UP**; STRONG_DOWN at the lower band becomes **CAPITULATION**.

Regime	Stance	Guidance (as shown)
Extended Up	LONG	Market extended. Momentum strong. Pullbacks are buying opportunities.
Strong Uptrend	LONG	Strong bullish conditions. Favor long setups with momentum.
Uptrend	LONG	Bullish but recovering. Favor long setups selectively.
Weak Uptrend	LONG	Recovering from weakness. Light long bias.
Pullback (bull intact)	LONG	Short-term pullback in a bull market. Watch for long re-entries.
Recovery	NEUTRAL	Recovering structure. Be selective. Wait for confirmation.
Basing	NEUTRAL	Market is basing. No clear direction. Reduce exposure.
Chop (bull bias)	NEUTRAL	Choppy but leaning bullish. Trade small and be selective.
Chop (bear bias)	NEUTRAL	Choppy with bear undertone. Reduce position size.
Correction (bull intact)	NEUTRAL	Correction in a bull market. Monitor for stabilization before re-entering.
Topping	SHORT	Signs of topping. Reduce longs. Hedge or go short on weakness.
Bear Rally	NEUTRAL	Counter-trend rally in bear market. Fade strength cautiously.
Downtrend	SHORT	Bearish conditions. Avoid longs. Short on rallies.
Strong Downtrend	SHORT	Strongly bearish. Reduce all risk. Short only.
Capitulation	SHORT	Market in capitulation. Extreme caution. Watch for reversal signals.

Confidence is high when long and mid term agree (both bullish-side or both bearish-side), medium otherwise; **Aligned / Diverging** says the same.

On each card: `regime`, `regimeLabel`, `longTerm`, `midTerm`, `shortTerm`.

4. Sector heatmap

15 ETFs: XLK Technology, XLF Financial, XLI Industrial, XLY Consumer Discretionary, XLE Energy, XLV Health Care, XLC Communication Services, XLU Utilities, XLB Materials, XLRE Real Estate, XLP Consumer Staples, SMH Semiconductors, IBB Biotechnology, XRT Retail, XTN Transportation.

Columns: Sector, ETF, Daily %, Week %, **RS Day** (ETF day − SPY day), **RS Week** (ETF week − SPY week), Bias, Score, ADX, Status (🔥 HOT). Sorted by score.

Score (`sectorShortTermBias`)

1. Five checks: close > SMA20 · SMA20 > SMA50 · close > VWAP · RS day > 0 · week > 0.
2. `base = checks / 5 × 100`.
3. `RS influence = RS day × 5`, clamped to ±30.
4. ADX factor: **1.3** if ADX > 25, **1.1** if 20–25, **0.8** below 20.
5. `score = round((base + RS influence) × ADX factor)`, clamped to ±100.
6. Bias: score > **20** BULLISH, < **−20** BEARISH (settings `sectorBullishThreshold`, `sectorBearishThreshold`), else NEUTRAL.

The score is lopsided: with no checks met it bottoms out around −39, while all five can reach +100 (F-16, open).

Hot (`updateHotSector`, per trading day)

- **Hot at once** when the score reaches **60** (`hotImmediateThreshold`).
- **Hot after 3 trading days** (`hotSustainedSessions`) whose best score reached **40** (`hotSustainedThreshold`).
- **Stays hot** while the day's best score is at least **20** (`hotFloorThreshold`). After **2** days below that (`coolOffDays`) it cools off.
- Counted per ET trading day, not per scan (F-17). The state lives in the tool's memory, so a restart starts it again; Settings has a reset.

On each card: `broadResolved` (the card's TradingView sector mapped to one of the 15), `secBias`, `secScore`, `secHot` (the 🔥 HOT badge). In R2: each sector's bias, score, change and hot, plus counts of bullish and bearish sectors.

5. L · Industry groups

qp ranks **industry groups across the whole market**, the level O'Neil says carries about 37 % of a stock's move (the sector carries 12 %).

Step	How
Membership	the industry map: **SIC codes from EDGAR** for every filer (TradingView labels only for symbols SIC has not classified, F-21)
Strength	each stock's RS rating from qp, ranked against the whole market
Group strength	the **median** member RS (one strong stock cannot carry a weak group)
Thin groups	a group with fewer than **6** ranked members (`QP_GROUP_MIN_MEMBERS`) is merged one rung up: 4-digit industry → 3-digit **industry group** → 2-digit **major group**; still too thin at the top, dropped
Ranking	groups ranked 1 … N by median RS; percentile and a letter (A+ ≥ 94 … E)
Rotation	rank three months ago → rank now: **into** / **out of** leadership
Refresh	nightly after the SIC pass; cached 12 hours; on request when missing

The table shows the **top 20** and the **weakest 10**:

- **Rank X of N**;
- **Group** name, marked *rolled up* when merged;
- **Pct** with its letter;
- **Members**;
- **Median RS**;
- **Over 80**, the share of members with RS over 80;
- **3mo**, the rank three months ago → now, coloured by rotation;
- **Leaders**, the top four members with their RS.

The note above the table gives how many symbols and industries are mapped, and how many groups have enough members to rank.

On each card: CANSLIM **L** (group, level, rank, RS inside its group, rotation) and the `csGroup*` columns in R1.

Who uses what on this tab

Number	Card	R1	R2	Model features
Regime, long/mid/short term	Market Context	✓	✓	✓
Sector bias, score, hot	Market Context, 🔥 HOT	✓	✓ (all 15)	bias and score (not hot)
Index prices	—	—	✓	—
O’Neil status	CANSLIM M	`csMarket`	✓ (read at export)	—
Group rank	CANSLIM L	`csGroup*`	—	—